

OpenAPI Mock Server Generator

Software Requirements Specification — Lab 1

Problem Context: A tool for API Developers and QA Engineers to validate OpenAPI specifications, generate mock servers, control their execution, and test API endpoints without requiring the actual backend.

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow an API Developer to upload or provide a valid OpenAPI specification in JSON or YAML format.	High	Valid OpenAPI JSON/YAML files are accepted; invalid or unsupported files are rejected with an error message.	The OpenAPI specification is the main input for generating the mock server.
FR-002	Functional	The system shall validate the uploaded OpenAPI specification and display validation errors when the specification is invalid.	High	For an invalid specification, the system identifies the validation error and displays it to the user.	Validation prevents incorrect mock APIs from being generated.
FR-003	Functional	The system shall generate a mock server containing endpoints defined in the valid OpenAPI specification.	High	For every valid API endpoint defined in the specification, the generated mock server provides a corresponding endpoint.	This is the core functionality of the system.
FR-004	Functional	The system shall allow the user to start, stop, and restart the generated mock server.	Medium	The user can start and stop the mock server, and the system displays its current status.	Developers and QA engineers need control over the mock server during API testing.
FR-005	Functional	The system shall allow the user to send test requests to generated API endpoints and display the corresponding mock responses.	High	When a valid request is sent to a generated endpoint, the system returns the configured mock response with the appropriate HTTP status code.	This allows QA engineers to test API integrations without requiring the actual backend.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The system shall generate and make a mock server available within 5 seconds for an OpenAPI specification containing up to 50 endpoints.	High	For a specification with ≤50 endpoints, at least 95 out of 100 generation attempts complete within 5 seconds.	Fast generation improves developer and QA productivity.
NFR-002	Nonfunctional	The system shall validate uploaded OpenAPI specifications and return a validation result within 2 seconds for specifications up to 1 MB in size.	Medium	At least 95 out of 100 validation requests for files ≤1 MB return a result within 2 seconds.	Quick feedback helps users identify and fix specification errors efficiently.

UML Use-Case Mapping

Actors: API Developer, QA Engineer

Use Cases: Upload OpenAPI Specification, Validate OpenAPI Specification, Generate Mock Server, Start/Stop/Restart Mock Server, Send Test Request & View Mock Response.

«**include**»: Generate Mock Server includes Validate OpenAPI Specification because validation is mandatory before generation.

«**extend**»: Validate OpenAPI Specification may be extended by Show Validation Errors when the specification is invalid; testing may be extended by Retry with Another

Request when required.